Summer 2026 Syllabus
2026 年夏季教学大纲

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Course Overview
课程简介

We study computational methods for solving macroeconomic models with dynamics, heterogeneous agents, and varying assumptions about expectations. We will do so using practical projects in which we will define macroeconomic models and then solve them in the programming language Julia. Starting with a simple consumption-savings model of a single agent, we will build up to representative-agent general equilibrium models, then heterogeneous-agent general equilibrium models. Finally, we will cover the "Continuation Value is All You Need" (CVIAYN) method for solving heterogeneous-agent dynamic models with uncertainty, as in Sun (2026).

我们研究求解宏观经济模型的计算方法，模型包含动态、异质性主体，以及对预期的不同假设。我们将通过实操项目进行：先定义宏观经济模型，再用编程语言 Julia 求解。从单一主体的简单消费–储蓄模型起步，逐步建立到代表性主体一般均衡模型，再到异质性主体一般均衡模型。最后，我们将介绍"续值即一切"（CVIAYN）方法，用于求解含不确定性的异质性主体动态模型，参见 Sun (2026)。

The course will consist of ten daily two-hour workshop-style lectures over the course of two weeks.

课程共十讲，每讲两小时，工作坊形式，分两周进行。

Course Materials
教学资料

No textbooks are required. I will provide code and lecture notes as the sole course materials.

不指定教材。我将提供代码和讲义作为唯一的课程资料。

Course Schedule
课程安排

Lecture / 讲次	**Description / 内容**
Lecture 1 第一讲 May 11	**Introduction.** Housekeeping. Overview of course. **Dynamic Modelling.** The consumption-savings problem in recursive form. **导论。**事务说明。课程概览。**动态建模。**递归形式下的消费–储蓄问题。
Lecture 2 第二讲 May 12	**Computation.** Introduction to Julia. Value function iteration. **计算。**Julia 简介。价值函数迭代。
Lecture 3 第三讲 May 13	**Heterogeneity.** Interpreting the grid-based solution to the consumption-savings problem as a solution for heterogeneous households. Solving the consumption-saving for agents who are heterogeneous in dimensions other than wealth. **异质性。**把消费–储蓄问题的网格解读作一群异质性家庭的解。求解在财富以外维度上具有异质性的主体的消费–储蓄问题。
Lecture 4 第四讲 May 14	**Equilibrium.** Heterogeneous Agents in Equilibrium: the Aiyagari model. Defining dynamic recursive equilibrium. **均衡。**均衡中的异质性主体：Aiyagari 模型。动态递归均衡的定义。
Lecture 5 第五讲 May 15	**Expectations.** Transition dynamics in the Aiyagari model. **预期。**Aiyagari 模型中的过渡动态。
Lecture 6 第六讲 May 18	**Uncertainty.** The Krusell-Smith Model. **不确定性。**Krusell–Smith 模型。
Lecture 7 第七讲 May 19	**Continuation Value is All You Need.** Solving the Krusell-Smith Model using neural networks. **续值即一切。**用神经网络求解 Krusell–Smith 模型。
Lecture 8 第八讲 May 20	**Richer Modelling.** The Stage Decomposition Method for building complex economic models. **更丰富的建模。**用阶段分解方法构建复杂的经济模型。
Lecture 9 第九讲 May 21	**Discrete Choice.** Incorporating discrete choice into a stage-decomposed model. Spatial Modelling. **离散选择。**在阶段分解模型中加入离散选择。空间建模。
Lecture 10 第十讲 May 22	**Final Steps.** Overview of applications. Discussion of estimation. **收官。**应用综览。关于估计的讨论。

Policies & Statements
政策与声明

Recording Lectures (by Student)
学生录制讲课

Students wishing to record lectures or other course material in any way are required to ask the instructor's explicit permission and may not do so unless permission is granted. This includes audio and video recording and photographing slides or other course materials.

如以任何方式录制课程或其他课程材料，须事先取得授课教师的明确许可；未获许可不得录制。这包括音频与视频录制，以及对幻灯片或其他课程材料的拍照。

Granting permission to record applies only for that individual student's own study purposes and does not include permission to "publish" or distribute them in any way. It is forbidden for a student to publish an instructor's notes on a website or to sell them in other form without formal permission.

获准录制仅适用于学生个人的学习目的，不包含以任何方式 "发布" 或传播的许可。学生不得将教师讲义发布到网站，或以其他形式出售，未经正式许可不得为之。

Equity, Diversity and Inclusion
公平、多元与包容

I am committed to equity and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect where all members of our community can express themselves, engage with each other, and respect one another's differences. I do not condone discrimination or harassment against any persons or communities.

我致力于推动公平并尊重多元。本课程学习环境中的每位成员都应努力营造相互尊重的氛围，使我们这个社区的所有成员都能表达自己、彼此交流，并尊重彼此的差异。我不容忍任何针对个人或群体的歧视或骚扰行为。